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Robustness of Information Circulation under Entropy Constraints

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Abstract

In complex systems, cyclic causal structures are closely associated with recurrent and oscillatory dynamics, yet an integrated dynamical description remains lacking. From an information-theoretic perspective, we formulate such structures as information circulation and show that, in finite systems, the presence of a cycle is necessary for bounded, nontrivial dynamics. To characterize the underlying mechanism of this behavior, we analyze the merging of information flows at a single node. Near the entropy-capacity limit, competing inputs become effectively exclusive: increasing the pre-existing circulating flow suppresses additional incoming information. This establishes an intrinsic robustness mechanism under finite entropy constraints. The mechanism is validated in Gaussian multiple-access channels and a multiplex-receiver model, while it is only partially satisfied for AND gate interactions. Simulations of a binary stochastic network with sigmoid response further confirm the predicted suppression of external inputs by internal circulation. These results identify information circulation as a structural and dynamical basis for robustness in finite complex systems.

1 Introduction

Many physical, biological, and engineered systems process several competing signals through components with finite local entropy capacity. For instance, when one pathway is already stabilized by cyclic feedback, a newly activated input competes with the pre-existing circulation under the entropy capacity of the receiving component. Although competition under finite capacity is intuitive and is closely related to optimization principles such as the information bottleneck [1, 2], it is not obvious how this competition is expressed dynamically at the level of directional information flow. This motivates a formulation in which cyclic causal relationships are described through information-dynamical quantities.

Cyclic causal relationships are widely observed in complex systems, including gene regulatory networks [3, 4, 5, 6, 7], thalamocortical brain circuits [8], Arctic climate feedbacks [9], and predator-prey ecosystems [10]. These structures can be interpreted as feedback loops that support directed information circulation and often exhibit oscillatory or recurrent behavior. However, a unified and statistically grounded description of such cyclic causal relationships remains lacking, particularly for analyzing their dynamical properties.

To address this issue, we adopt an information-theoretic framework, which provides a model-independent and universally applicable description of correlations and causality in probabilistic systems. Within this framework, cyclic causal relationships can be formulated as cycles in an information dynamical network.

Information dynamics has been extensively developed across multiple disciplines, such as information thermodynamics [11, 12, 13, 14, 15], quantum thermodynamics [16, 17, 18], neuroscience [19, 20], and bioinformatics [21]. These approaches describe entropy production in non-equilibrium systems and enable the identification of causal interactions in complex networks of neurons, genes, and proteins. In particular, mutual information and transfer entropy have been widely used to characterize correlations and directional information flow in many-body systems [22, 23, 24, 25, 13].

Building on these developments, Ref. [26] established a general and exact framework for information dynamics in which individual entropy, mutual information, transfer entropy, and

reversed transfer entropy [27, 28, 15] are treated as the dynamical variables governed by a set of evolution equations with control variables. In that framework, the zero-control limit was analyzed, revealing that nonzero control variables induce curvature in the evolution of these dynamical variables and reflect the statistical structure of the underlying system.

Despite its generality, this framework does not explicitly address interactions between information flows, which are necessary to describe the dynamical behavior and responses of systems with cyclic structures.

In this paper, we extend the framework of Ref. [26] to describe cyclic causal relationships as cycles of transfer entropy, which encode directional causality [29]. Under locally deterministic dynamics and non-synergistic interactions, we demonstrate in Sec. 3 that a finite directed network of transfer entropy must contain a cycle in order to exhibit bounded and nontrivial dynamics. This result identifies information circulation as a necessary structural feature of such dynamics under the present assumptions.

To further characterize these dynamics, Sec. 4 analyzes the behavior of a single cycle within the information dynamical framework. While the stationary state of an isolated cycle was previously studied in Ref. [26], we introduce interactions between the cycle and an external information source to investigate how such interactions modify the dynamics.

The analysis reveals that the behavior of the system depends critically on how control variables are assigned at points where multiple information flows merge. Accordingly, Sec. 5 examines the information dynamics at such merging points. Under a set of assumptions, we show that competing information flows become effectively exclusive when the individual entropy at the merging node is constrained, indicating a form of robustness of the information circulation.

2 Information Dynamical Equations

2.1 Definitions

We summarize the information-theoretic quantities with increasing level of structure, from pairwise measures to directional and multi-variable extensions.

Definition 1 The *Shannon entropy* $H(X)$ and *conditional entropy* $H(Y|X)$ [30, 31] are defined as:

$$\begin{aligned} H(X) &:= - \sum_{x \in \mathcal{M}} P(X = x) \ln P(X = x) \\ H(Y|X) &:= H(X, Y) - H(X), \end{aligned} \quad (1)$$

where X and Y are discrete random variables taking values in the state space $\mathcal{M}$.

Also, the entropy of a set of random variables $V = \{V_1, \dots, V_m\}$ is defined with the joint probability distribution:

$$H(V) := - \sum_{v_1, \dots, v_m \in \mathcal{M}} P(V_1 = v_1, \dots, V_m = v_m) \ln P(V_1 = v_1, \dots, V_m = v_m). \quad (2)$$

Definition 2 The *mutual information* between X and Y is defined as [31]:

$$I(X, Y) := H(X) + H(Y) - H(X, Y), \quad (3)$$

and the *conditional mutual information* is given by

$$I(X, Y|Z) := H(X|Z) + H(Y|Z) - H(X, Y|Z). \quad (4)$$

Definition 3 The *transfer entropy* [32, 33] from Y to X , which quantifies directional information flow, is defined as:

$$T_{Y \rightarrow X} := I(X', Y|X) \quad (5)$$

where X and X' are evaluated at time $t = t_0$ and $t = t_1$, respectively, with $t_1 > t_0$.

Plus, all primed variables in this paper are evaluated one time step after the indicated base time. For instance, X'' is evaluated at time $t = t_2$.

Definition 4 The *reversed transfer entropy* [27, 28, 15] (also referred to as backward transfer entropy [27]) is defined as,

$$T_{Y \rightarrow X}^\dagger := I(X, Y'|X'), \quad (6)$$

which corresponds to the transfer entropy evaluated under time-reversed indexing. All quantities defined above are non-negative.

Definition 5 The **co-information** (also called interaction information or multiple mutual information) [34, 35, 36, 37, 38, 39, 40, 41] captures multi-correlations, such as synergy and redundancy, and is defined as:

$$I(V) := \sum_{i=1}^{|V|} (-1)^{i+1} \sum_{U \subset V, |U|=i} H(U). \quad (7)$$

Similarly, the **conditional co-information** is,

$$I(V|W) := \sum_{i=1}^{|V|} (-1)^{i+1} \sum_{U \subset V, |U|=i} H(U|W). \quad (8)$$

Definition 6 The **multiple transfer entropy** and **multiple reversed transfer entropy** [26] are defined as,

$$\begin{aligned} T_{V \rightarrow U} &:= I(V \cup U' | U) \\ T_{V \rightarrow U}^\dagger &:= I(V' \cup U | U'). \end{aligned} \quad (9)$$

2.2 The General Framework of Information Dynamics

For a general multi-variable system with N components, we formulate the dynamical equations[26] governing the temporal evolution of the information-theoretic quantities defined in Sec. 2.1. In this framework, the entropy, mutual information, transfer entropy, and reversed transfer entropy are treated as state variables whose temporal changes are described by a set of coupled evolution equations.

Let $\Gamma_1, \dots, \Gamma_N$ denote the random variables associated with N components and $\Delta H(\Gamma_i)$, $\Delta I(\Gamma_i, \Gamma_j)$, $\Delta T_{\Gamma_j \rightarrow \Gamma_i}$, and $\Delta T_{\Gamma_j \rightarrow \Gamma_i}^\dagger$ denote the temporal increments of the corresponding quantities over a discrete time step.

$$\begin{cases} \Delta H(\Gamma_i) = \sum_{k \in K_i} [T_{k \rightarrow \Gamma_i} - T_{k \rightarrow \Gamma_i}^\dagger] - E_{K_i \rightarrow \Gamma_i} + \alpha_1^{K_i \rightarrow \Gamma_i} \\ \Delta I(\Gamma_i, \Gamma_j) = [T_{\Gamma_j \rightarrow \Gamma_i} - T_{\Gamma_j \rightarrow \Gamma_i}^\dagger] + [T_{\Gamma_i \rightarrow \Gamma_j} - T_{\Gamma_i \rightarrow \Gamma_j}^\dagger] + \alpha_2^{\Gamma_i, \Gamma_j} \\ \Delta T_{\Gamma_j \rightarrow \Gamma_i} = \sum_{k \in K_j - \{\Gamma_i\}} [T_{k \rightarrow \Gamma_j} - T_{k \rightarrow \Gamma_j}^\dagger] - [T_{\Gamma_j \rightarrow \Gamma_i} - T_{\Gamma_j \rightarrow \Gamma_i}^\dagger] - E_{K_j \rightarrow \Gamma_j} + \alpha_1^{K_j \rightarrow \Gamma_j} - \alpha_2^{\Gamma_i, \Gamma_j} - \alpha_3^{\Gamma_j \rightarrow \Gamma_i} \\ \Delta T_{\Gamma_j \rightarrow \Gamma_i}^\dagger = \sum_{k \in K_j - \{\Gamma_i\}} [T_{k \rightarrow \Gamma_j} - T_{k \rightarrow \Gamma_j}^\dagger] - [T_{\Gamma_j \rightarrow \Gamma_i} - T_{\Gamma_j \rightarrow \Gamma_i}^\dagger] - E_{K_j \rightarrow \Gamma_j} + \alpha_1^{K_j \rightarrow \Gamma_j} - \alpha_2^{\Gamma_i, \Gamma_j} \\ + \alpha_4^{\Gamma_j} - \alpha_5^{\Gamma_i, \Gamma_j} - \alpha_6^{\Gamma_j \rightarrow \Gamma_i}, \end{cases} \quad (10)$$

with,

$$\begin{aligned} \alpha_1^{K_i \rightarrow \Gamma_i} &:= H(\Gamma'_i | \Gamma_i, K_i) - H(\Gamma_i | \Gamma'_i, K'_i) \\ \alpha_2^{\Gamma_i, \Gamma_j} &:= I(\Gamma'_i, \Gamma'_j | \Gamma_i, \Gamma_j) - I(\Gamma_i, \Gamma_j | \Gamma'_i, \Gamma'_j) \\ \alpha_3^{\Gamma_j \rightarrow \Gamma_i} &:= \Delta H(\Gamma_j | \Gamma_i, \Gamma'_i) \\ \alpha_4^{\Gamma_j} &:= \Delta^2 H(\Gamma_j) = \Delta(\Delta H(\Gamma_j)) = H(\Gamma''_j) - 2H(\Gamma'_j) + H(\Gamma_j) \\ \alpha_5^{\Gamma_i, \Gamma_j} &:= \Delta^2 I(\Gamma_i, \Gamma_j) = I(\Gamma''_i, \Gamma''_j) - 2I(\Gamma'_i, \Gamma'_j) + I(\Gamma_i, \Gamma_j) \\ \alpha_6^{\Gamma_j \rightarrow \Gamma_i} &:= \Delta H(\Gamma'_j | \Gamma_i, \Gamma'_i). \end{aligned} \quad (11)$$

Here, K_i is a set of causal parents of component i and $K'_i := \{k' | k \in K_i\}$.

The term $E_{K_i \rightarrow \Gamma_i}$ is responsible for the coupling between first-order(pairwise) correlations and

high-order correlations: for $V, U, W \subset \{\Gamma_1, \dots, \Gamma_N\}$,

$$\left\{ \begin{array}{l} E_{K_i \rightarrow \Gamma_i} := \sum_{u=2}^{|K_i|} (-1)^u \sum_{V \subset K_i, |V|=u} [T_{V \rightarrow \Gamma_i} - T_{V \rightarrow \Gamma_i}^\dagger] \\ \Delta I(V) = \sum_{(U, W) \in D_V} [T_{U \rightarrow W} - T_{U \rightarrow W}^\dagger] + \alpha_2^V \\ \Delta T_{U \rightarrow W} = (-1)^{|W|} \Delta I(U \cup W) + \sum_{u=1}^{|W|} (-1)^{|W|-u} \sum_{S \subset W, |S|=u} (\Delta I(U \cup (W \setminus S))) \\ \quad + \sum_{s=1}^u (-1)^s \sum_{Z \subset S, |Z|=s} \alpha_3^{U \cup (W \setminus S) \rightarrow Z} \\ \Delta T_{U \rightarrow W}^\dagger = (-1)^{|W|} \Delta I(U \cup W) + (-1)^{|W|} \alpha_5^{U \cup W} + \sum_{u=1}^{|W|} (-1)^{|W|-u} \sum_{S \subset W, |S|=u} (\Delta I(U \cup (W \setminus S))) \\ \quad + \alpha_5^{U \cup (W \setminus S)} + \sum_{s=1}^u (-1)^s \sum_{Z \subset S, |Z|=s} \alpha_6^{U \cup (W \setminus S) \rightarrow Z}, \end{array} \right. \quad (12)$$

with,

$$\begin{aligned} \alpha_2^V &:= I(V'|V) - I(V|V') \\ \alpha_3^{X \rightarrow Z} &:= \Delta I(X|Z \cup Z') \\ \alpha_5^V &:= \Delta^2 I(V) \\ \alpha_6^{X \rightarrow Z} &:= \Delta I(X'|Z \cup Z'). \end{aligned} \quad (13)$$

Each equation is exact [26] and consists of an intrinsic contribution, determined by the probabilistic structure of the system, and a control contribution, parameterized by coefficients α , which encode deviations from the zero-control limit. In the zero-control limit, these equations reduce to the intrinsic dynamics analyzed in Ref. [26]. Furthermore, the control variables α encode how interaction structures and constraints modify the intrinsic information dynamics. Their values are determined by mechanisms such as blocked channels and finite-capacity effects at merging points.

Ref. [26] demonstrated that the non-trivial circulation near the zero-control limit arises from the restriction allowing the one-way interaction through one channel between component A and B :

$$\begin{cases} [T_{B \rightarrow A} = 0 \text{ and } T_{A \rightarrow B}^\dagger = 0] \\ \text{or } [T_{A \rightarrow B} = 0 \text{ and } T_{B \rightarrow A}^\dagger = 0]. \end{cases} \quad \text{for all time} \quad (14)$$

2.3 Structural Assumption : Deterministic Reversible Dynamics

The analysis of Sec. 3 assumes the deterministic reversible dynamics: $\exists U = (U_1, U_2, \dots, U_N)$ and U^{-1} such that,

$$\begin{aligned} \Gamma'_i &= U_i(\Gamma_1, \dots, \Gamma_N, t_0) \\ \Gamma_i &= U_i^{-1}(\Gamma'_1, \dots, \Gamma'_N, t_1), \end{aligned} \quad (15)$$

where $i \in \{1, 2, \dots, N\}$ and $t_1 > t_0$. Then, with $K_i^{total} := \{\Gamma_1, \dots, \Gamma_N\} - \{\Gamma_i\}$, the following uncertainty vanishes:

$$\begin{aligned} H(\Gamma'_i | \Gamma_i, K_i^{total}) &= 0 \\ H(\Gamma_i | \Gamma'_i, K_i^{total}) &= 0. \end{aligned} \quad (16)$$

Hence, the control parameter $\alpha_1^{K_i^{total} \rightarrow \Gamma_i}$ defined in the equation (11) becomes zero in this case:

$$\alpha_1^{K_i^{total} \rightarrow \Gamma_i} := H(\Gamma'_i | \Gamma_i, K_i^{total}) - H(\Gamma_i | \Gamma'_i, K_i^{total}) = 0. \quad (17)$$

Furthermore, if one assumes the locally deterministic reversible dynamics, the set K_i can be effectively shrunk to the set of the causal parents: $K_i := \{k_1, \dots, k_m\} \subset \{\Gamma_1, \dots, \Gamma_N\}$, such that,

$$\begin{aligned} \Gamma'_i &= U_i(\Gamma_i, k_1, \dots, k_m) \\ \Gamma_i &= U_i^{-1}(\Gamma'_i, k'_1, \dots, k'_m). \end{aligned} \quad (18)$$

Therefore,

$$\alpha_1^{K_i \rightarrow \Gamma_i} := H(\Gamma'_i | \Gamma_i, K_i) - H(\Gamma_i | \Gamma'_i, K'_i) = 0. \quad (19)$$

2.4 Structural Assumption : Absence of Synergistic Interactions

Let the neighbor set of a component v be $K = \{k_1, \dots, k_m\}$, and define an arbitrary but fixed tuple $\mathbf{K} := (k)_{k \in K}$. Then, the transfer entropy and reversed transfer entropy,

$$\begin{aligned} T_{\mathbf{K} \rightarrow v} &:= I(\mathbf{K}, v' | v) \\ T_{\mathbf{K} \rightarrow v}^\dagger &:= I(\mathbf{K}', v | v'), \end{aligned} \quad (20)$$

are distinct from the multiple transfer entropy $T_{K \rightarrow \{v\}}$ and multiple reversed transfer entropy $T_{K \rightarrow \{v\}}^\dagger$ defined in Section 2.1.

Decomposing $T_{\mathbf{K} \rightarrow v}$ and $T_{\mathbf{K} \rightarrow v}^\dagger$ yields:

$$\begin{aligned} T_{\mathbf{K} \rightarrow v} &= T_{k \rightarrow v} + I(\mathbf{K}_{\setminus k}, v' | v, k) \\ T_{\mathbf{K} \rightarrow v}^\dagger &= T_{k \rightarrow v}^\dagger + I(\mathbf{K}'_{\setminus k}, v | v', k'), \end{aligned} \quad (21)$$

for $k \in K$.

If $T_{\mathbf{K} \rightarrow v} = 0$ and $T_{\mathbf{K} \rightarrow v}^\dagger = 0$, then $T_{k \rightarrow v} = 0$ and $T_{k \rightarrow v}^\dagger = 0$ for all $k \in K$, since the right hand terms are non-negative. Nevertheless, the converse is not generally true, as the counterexample is presented in Appendix A. Restricting to non-synergistic interaction, the analysis of Sec. 3 assumes that the converse holds, i.e., if $T_{k \rightarrow v} = 0$ and $T_{k \rightarrow v}^\dagger = 0$ for all $k \in K$, then $T_{\mathbf{K} \rightarrow v} = 0$ and $T_{\mathbf{K} \rightarrow v}^\dagger = 0$. This assumption excludes the pure synergistic effect, such as the XOR operation (Appendix A), while it allows the cases of Gaussian MAC (Sec. 5.4.1) and additive sigmoid (Hill-type) response functions (Sec. 6).

3 General Information Dynamics of a Finite System

We consider an N -component system satisfying the blocking flows condition (Eq. (14)) and the assumption of deterministic reversible dynamics (Sec. 2.3). The system can be represented as a directed finite network, where directed edges correspond to nonzero pairwise transfer entropy (or reversed transfer entropy) from fixed causal parents satisfying Eq. (18). To analyze the role of network structure, we distinguish between acyclic networks and networks containing at least one cycle. We recall the following definitions from graph theory [42, 43].

Definition 7 A **directed graph** is a pair $G = (V, \vec{E})$, where V is a finite set of vertices and $\vec{E} \subset \{(x, y) \in V \times V | x \neq y\}$ is a set of directed edges. Also, the reversed edge set $\overleftarrow{E} := \{(y, x) | (x, y) \in \vec{E}\}$ is defined. Then, the **in-degree** of a vertex $v \in V$ is $d^-(v) = |\{x \in V | (x, v) \in \vec{E}\}|$, while the **out-degree** of a vertex $v \in V$ is $d^+(v) = |\{x \in V | (v, x) \in \vec{E}\}|$.

Throughout this section, a directed graph $G = (V, \vec{E})$ is assumed to have no isolated vertices; that is, every vertex $v \in V$ satisfies $d^-(v) + d^+(v) > 0$.

Definition 8 A **source** in a directed graph $G = (V, \vec{E})$ is the vertex $v \in V$ with $d^-(v) = 0$ and $d^+(v) \neq 0$, and a **sink** in a directed graph $G = (V, \vec{E})$ is the vertex $v \in V$ with $d^+(v) = 0$ and $d^-(v) \neq 0$.

Definition 9 A **cycle** C in a directed graph $(V, \vec{E})$ is a sequence of distinct vertices $v_1, \dots, v_n$, with $n \geq 2$, such that $(v_i, v_{i+1}) \in \vec{E}$ for $i = 1, \dots, n-1$, and $(v_n, v_1) \in \vec{E}$.

Definition 10 A directed graph is **acyclic** if it contains no cycle.

For each component v , let $K_v \subset V \setminus \{v\}$ denote the set of causal parents of v , namely the components whose states enter the local update rule of v as Eq. (18). Then, a directed graph of pairwise transfer entropy is defined as a directed graph $G_T = (V, \vec{E})$ with $\vec{E} = \{(x, y) \in V \times V | x \neq y, x \in K_y, T_{x \rightarrow y} > 0\}$.

The Proposition 1 and Proposition 2 show that acyclic information networks necessarily exhibit monotonic entropy evolution, preventing fully recurrent or stationary nontrivial dynamics under the present assumptions.

Proposition 1 *An acyclic graph must include at least one source and sink.*

(proof) From Proposition (3.19) in the chapter "Directed Acyclic Graphs and Topological Ordering" of Ref. [43], a finite acyclic graph $G = (V, \vec{E})$ must include at least one source. Since the directed graph with reversed edges $G^{rev} = (V, \overleftarrow{E})$ is also acyclic, it has a source w that is the sink in the original graph $G = (V, \vec{E})$. ■

While Proposition 1 is a classical result of graph theory, Proposition 2 focuses on the information dynamical aspect under the framework of 2.2, implying that acyclic information networks cannot sustain nontrivial dynamical behavior.

Proposition 2 *Under the standing assumptions of this section and the non-synergistic assumption of Sec. 2.4, if the directed network of pairwise transfer entropy is acyclic, then the individual entropy of a sink (or source) evolves monotonically in time.*

(proof) From Proposition 1, a directed graph of pairwise transfer entropy $G_T = (V, \vec{E})$ has a sink $v \in V$: $T_{k \rightarrow v} > 0$ and $T_{v \rightarrow k} = 0$ for all $k \in K$, where the set of v 's neighbors in G_T is $K := \{k \in V | (k, v) \in \vec{E}\}$. Then, the dynamics of v 's individual entropy (Eq. (10)) is,

$$\begin{aligned} \Delta H(v) &= \sum_{k \in K} [T_{k \rightarrow v} - T_{k \rightarrow v}^\dagger] - E_{K \rightarrow v} + \alpha_1^{K \rightarrow v} \\ &= \sum_{k \in K} [T_{k \rightarrow v} - T_{k \rightarrow v}^\dagger] - E_{K \rightarrow v} \\ &= T_{\mathbf{K} \rightarrow v} - T_{\mathbf{K} \rightarrow v}^\dagger, \end{aligned} \quad (22)$$

where $\mathbf{K} := (k)_{k \in K}$.

Here, $\alpha_1^{K \rightarrow v} = 0$ due to the assumption of deterministic and reversible dynamics.

Since $T_{k \rightarrow v}^\dagger = 0$ for all $k \in K$ from the blocking flows condition, the assumption of Section 2.4 requires $T_{\mathbf{K} \rightarrow v}^\dagger = 0$. In addition, by definition, $T_{\mathbf{K} \rightarrow v} := I(\mathbf{K}, v' | v) \geq 0$. If $T_{\mathbf{K} \rightarrow v} = 0$, then $T_{k \rightarrow v} = 0$ for all $k \in K$ by the argument in Section 2.4, which contradicts the fact that v is the sink.

Therefore,

$$\Delta H(v) = T_{\mathbf{K} \rightarrow v} > 0. \quad (23)$$

For the case where the vertex v is a source, $\Delta H(v) = -T_{\mathbf{K} \rightarrow v}^\dagger < 0$, indicating $H(v)$ decreases monotonically over time. ■

Such monotonic behavior corresponds to trivial dynamics, in the sense that the system cannot sustain oscillatory or stationary nontrivial states. If the entropy $H(v)$ is bounded, it must converge to a constant value. If unbounded, the dynamics cannot form an attractor, which is necessary for nontrivial behaviors, such as oscillations [26] or relaxation to a stable point. Therefore, the presence of a cycle is a necessary condition for nontrivial dynamics in finite information networks.

4 Information Dynamics of a Finite System with a Cycle

In contrast to acyclic networks, finite systems containing a cycle can exhibit nontrivial dynamics. The information dynamics of such systems is analyzed within the framework of Sec. 2, focusing on the behavior of a single cycle and its interaction with an external information source. Prior to that, the dynamics of an isolated cycle is considered.

4.1 The Stationary State of a Single Cycle

Let an N -component system $\Gamma = \{\Gamma_0, \Gamma_1, \dots, \Gamma_{N-1}\}$ form a one-dimensional system with periodic boundary condition. In the index notation, $\text{mod } N$ is assumed in this section. By imposing the blocking flows condition and near the zero-control limit, Ref. [26] obtains,

$$\begin{aligned} \Delta H(\Gamma_i) &= T_{\Gamma_{i-1} \rightarrow \Gamma_i} - T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger \\ \Delta I(\Gamma_i, \Gamma_{i+1}) &= T_{\Gamma_i \rightarrow \Gamma_{i+1}} - T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger \\ \Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}} &= T_{\Gamma_{i-1} \rightarrow \Gamma_i} - T_{\Gamma_i \rightarrow \Gamma_{i+1}} \\ \Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}}^\dagger &= 0 \\ \Delta T_{\Gamma_i \rightarrow \Gamma_{i-1}} &= 0 \\ \Delta T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger &= T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger - T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger. \end{aligned} \quad (24)$$

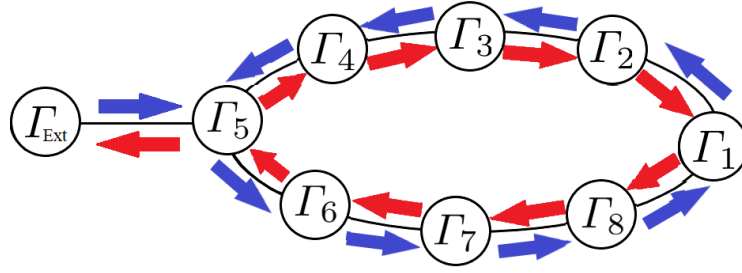


Figure 1: Schematic illustrations of information dynamics on a cycle interacting with a random source (*Ext*). Blue arrows represent transfer entropy, and red arrows represent reversed transfer entropy.

These equations describe the circulation of transfer entropy and reversed transfer entropy along the cycle. A stationary state is achieved when the transfer entropy and reversed transfer entropy are uniformly distributed along the cycle, satisfying $T_{\Gamma_{i-1} \rightarrow \Gamma_i} = T_{\Gamma_i \rightarrow \Gamma_{i+1}}$, $T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger = T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger$, and $T_{\Gamma_{i-1} \rightarrow \Gamma_i} = T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger$. This corresponds to a constant circulation of information along the cycle. As shown in Ref. [44], small deviations from this stationary state remain bounded in the analyzed examples, suggesting the stability of the circulating information flow.

4.2 The Transient Behaviors of the Single Cycle and a Random Source

Consider the interaction between a single cycle and an external information source. Let an external component *Ext* begin to interact with a component p in the cycle at the time $t = t_1$ as shown in Fig. 1, while no interaction is present at earlier times. Therefore, the stationary state of the previous section is assumed for the time $t = t_0 < t_1$. Then, the resulting dynamics is described by Proposition 3.

Proposition 3

$$\begin{aligned}
\Delta H(\Gamma_i) &= T_{\Gamma_{i-1} \rightarrow \Gamma_i} - T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger \quad (i \neq p) \\
\Delta H(\Gamma_p) &= T_{\Gamma_{p-1} \rightarrow \Gamma_p} + T_{\Gamma_{Ext} \rightarrow \Gamma_p} - T_{\Gamma_{p+1} \rightarrow \Gamma_p}^\dagger - E_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} \\
\Delta I(\Gamma_i, \Gamma_{i+1}) &= T_{\Gamma_i \rightarrow \Gamma_{i+1}} - T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger \\
\Delta I(\Gamma_p, \Gamma_{Ext}) &= T_{\Gamma_{Ext} \rightarrow \Gamma_p} - T_{\Gamma_p \rightarrow \Gamma_{Ext}}^\dagger \\
\\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}} &= T_{\Gamma_{i-1} \rightarrow \Gamma_i} - T_{\Gamma_i \rightarrow \Gamma_{i+1}} \quad (i \neq p-1, p) \\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}}^\dagger &= 0 \\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i-1}} &= 0 \\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger &= T_{\Gamma_{i-1} \rightarrow \Gamma_{i-2}}^\dagger - T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger \quad (i \neq p, p+1) \\
\\
\Delta T_{\Gamma_{p-1} \rightarrow \Gamma_p} &= T_{\Gamma_{p-2} \rightarrow \Gamma_{p-1}} - T_{\Gamma_{p-1} \rightarrow \Gamma_p} - \alpha_3^{\Gamma_{p-1} \rightarrow \Gamma_p} \\
\Delta T_{\Gamma_p \rightarrow \Gamma_{p-1}}^\dagger &= T_{\Gamma_{p-1} \rightarrow \Gamma_{p-2}}^\dagger - T_{\Gamma_p \rightarrow \Gamma_{p-1}}^\dagger - E'_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} \\
\Delta T_{\Gamma_p \rightarrow \Gamma_{p+1}} &= T_{\Gamma_{p-1} \rightarrow \Gamma_p} + T_{\Gamma_{Ext} \rightarrow \Gamma_p} - T_{\Gamma_p \rightarrow \Gamma_{p+1}} - E_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} \\
\Delta T_{\Gamma_{p+1} \rightarrow \Gamma_p}^\dagger &= \alpha_1^{\{\Gamma_p\} \rightarrow \Gamma_{Ext}} - \alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p} + T_{\Gamma_p \rightarrow \Gamma_{p-1}}^\dagger - T_{\Gamma_{p+1} \rightarrow \Gamma_p}^\dagger \\
&\quad - E'_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} - \alpha_6^{\Gamma_p \rightarrow \Gamma_{p-1}} \\
\\
\Delta T_{\Gamma_{Ext} \rightarrow \Gamma_p} &= \alpha_1^{\{\Gamma_p\} \rightarrow \Gamma_{Ext}} - T_{\Gamma_{Ext} \rightarrow \Gamma_p} - \alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p} \\
\Delta T_{\Gamma_p \rightarrow \Gamma_{Ext}}^\dagger &= -\alpha_{3456}^{\Gamma_p \rightarrow \Gamma_{Ext}}
\end{aligned} \tag{25}$$

(proof) Appendix B.

These equations show that the external source modifies the information flow locally at component p , where the incoming flow $T_{\Gamma_{Ext} \rightarrow \Gamma_p}$ merges with the circulating flow $T_{\Gamma_{p-1} \rightarrow \Gamma_p}$. The corresponding control variables determine how the newly activated interaction is incorporated into the pre-existing circulation.

Specifically, the terms, $\alpha_3^{\Gamma_{p-1} \rightarrow \Gamma_p}$, $\alpha_6^{\Gamma_p \rightarrow \Gamma_{p-1}}$, $\alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p}$, and $\alpha_6^{\Gamma_p \rightarrow \Gamma_{Ext}}$, are expected to become nonzero during the transient period after $t = t_1$, reflecting the local reorganization of information flow at the merging component p . Since the transfer entropy $T_{\Gamma_{Ext} \rightarrow \Gamma_p} = 0$ for $t = t_0 < t_1$, the control variable $\alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p}$ determines the emergence of the nonzero transfer entropy $T_{\Gamma_{Ext} \rightarrow \Gamma_p}$ at later times. This interaction can therefore be interpreted as the merging of two information flows, $T_{\Gamma_{Ext} \rightarrow \Gamma_p}$ and $T_{\Gamma_{p-1} \rightarrow \Gamma_p}$, at the component p . The control variables $\alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p}$ and $\alpha_3^{\Gamma_{p-1} \rightarrow \Gamma_p}$ govern the dynamics of this merging process.

5 The Merging of Two Information Flows

When a cycle of information flow interacts with an external information source, the manner in which two information flows merge at a single component determines the resulting dynamics. The merging of two information flows is analyzed within the framework of Sec. 2.2.

5.1 Problem Setting

Let A, B, P and Q be random variables defined over an ensemble, representing the states of four components. We consider a configuration in which the components represented by A and B interact with the component represented by P , while P also interacts with the component represented by Q (Fig. 2). At time $t = t_0$, the components A, P , and Q interact with each other, with the non-zero stationary transfer entropy $T_{A \rightarrow P} = T_{P \rightarrow Q} = \mu_A \neq 0$ and reversed transfer entropy

$T_{P \rightarrow A}^\dagger = T_{Q \rightarrow P}^\dagger = \mu_A$, while the component B is not interacting with the component P , implying $T_{B \rightarrow P} = 0$ and $T_{P \rightarrow B}^\dagger = 0$. Then, the components B and P start to interact at the time $t = t_1$. The quantities $T_{B' \rightarrow P'}$ and $T_{P' \rightarrow B'}^\dagger$ are determined from the dynamical equations (Eq. (10)).

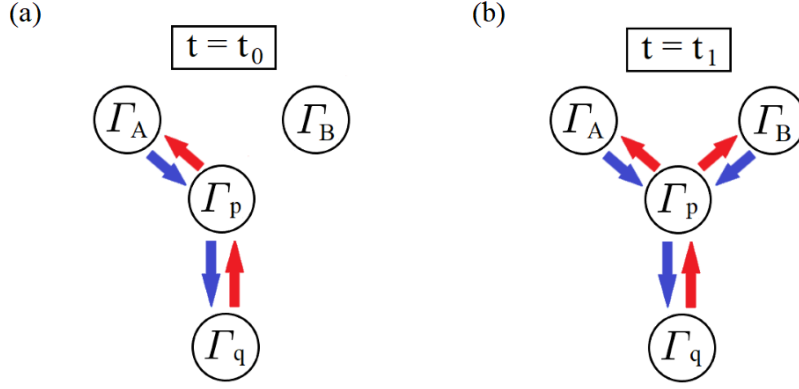


Figure 2: Schematic illustration of the merging of two information flows at time **(a)** $t = t_0$ and **(b)** $t = t_1$. Blue arrows denote transfer entropy, and red arrows denote reversed transfer entropy.

$$\begin{aligned}
 T_{B' \rightarrow P'} &= \Delta T_{B \rightarrow P} = \sum_{k \in K_B \setminus \{P\}} [T_{k \rightarrow B} - T_{k \rightarrow B}^\dagger] \\
 &\quad - E_{K_B \rightarrow B} + \alpha_1^{K_B \rightarrow B} - \alpha_2^{B, P} - \alpha_3^{B \rightarrow P}, \\
 T_{P' \rightarrow B'} &= \Delta T_{P \rightarrow B}^\dagger \\
 &= [T_{A \rightarrow P} - T_{Q \rightarrow P}^\dagger] - E_{\{A, B, Q\} \rightarrow P} \\
 &\quad + \alpha_1^{\{A, B, Q\} \rightarrow P} - \alpha_2^{B, P} + \alpha_4^P - \alpha_5^{B, P} - \alpha_6^{P \rightarrow B}.
 \end{aligned} \tag{26}$$

The conditions $T_{B \rightarrow P} = T_{P \rightarrow B}^\dagger = 0$ and the stationary conditions $\Delta I(B, P) = 0$ imply that $\alpha_2^{B, P} = 0$. Plus, $E_{\{A, B, Q\} \rightarrow P} = E_{\{A, Q\} \rightarrow P} = 0$, as the component B does not interact with component P at time $t = t_0$.

$$\begin{aligned}
 T_{B' \rightarrow P'} &= \sum_{k \in K_B \setminus \{P\}} [T_{k \rightarrow B} - T_{k \rightarrow B}^\dagger] - E_{K_B \rightarrow B} \\
 &\quad + \alpha_1^{K_B \rightarrow B} - \alpha_3^{B \rightarrow P}, \\
 T_{P' \rightarrow B'} &= [T_{A \rightarrow P} - T_{Q \rightarrow P}^\dagger] + \alpha_1^{\{A, B, Q\} \rightarrow P} \\
 &\quad + \alpha_4^P - \alpha_5^{B, P} - \alpha_6^{P \rightarrow B}.
 \end{aligned} \tag{27}$$

5.2 The Zero-control Limit

The structural role of control variables α in the merge problem is recognized by analyzing the uncontrolled behavior of dynamical variables. In the zero-control limit with the blocking flow condition, all control variables vanish except those associated with blocked channels. Then, the equation (27) in the limit becomes,

$$T_{B' \rightarrow P'} = \sum_{k \in K_B \setminus \{P\}} [T_{k \rightarrow B} - T_{k \rightarrow B}^\dagger] - E_{K_B \rightarrow B}. \tag{28}$$

The transfer entropy $T_{B' \rightarrow P'}$ is determined solely by incoming information flows and is not constrained by the entropy $H(P)$. Since $T_{A' \rightarrow P'} = T_{A \rightarrow P} = \mu_A$ and $T_{Q' \rightarrow P'}^\dagger = T_{Q \rightarrow P}^\dagger = \mu_A$ by the zero-control limit,

$$\Delta H(P') = [T_{A' \rightarrow P'} + T_{B' \rightarrow P'} - T_{Q' \rightarrow P'}^\dagger] = T_{B' \rightarrow P'}. \tag{29}$$

Consequently, the entropy $H(P')$ can increase without bound, indicating the absence of a stationary state with finite entropy. Nevertheless, it is not a general case. For instance, if the component P has a binary state, then its entropy $H(P') \leq \ln 2$. In summary, the zero-control limit allows an unlimited increase of the component P 's entropy $H(P')$, as the determination of transfer entropy $T_{B' \rightarrow P'}$ is independent of the entropy $H(P')$. Hence, the non-zero control variables α are required to introduce a finite maximum entropy of the component P , the merging point.

5.3 Near the Zero-control Limit with a fixed maximum entropy

A finite upper bound on the entropy $H(P') \leq \ln Q$ imposes constraints on the merging dynamics. The regime $H(P') \rightarrow \ln Q$ is examined, under the following assumptions:

$$\begin{aligned}
&\text{Assumption 1 : } T_{A' \rightarrow P'} - T_{\{A', B'\} \rightarrow P'} = I(A', P'' | P', B') = T_{A \rightarrow P} \\
&\text{Assumption 2 : } I(Q, P' | P, A) = 0, \quad I(Q', P'' | P', A', B') = 0 \\
&\text{Assumption 3 : } \alpha_1^{\{A, Q\} \rightarrow P} = \alpha_1^{\{A', B', Q'\} \rightarrow P'} \\
&\text{Assumption 4 : } \frac{\partial \alpha_1^{\{A, Q\} \rightarrow P}}{\partial T_{A \rightarrow P}} > -1 \\
&\text{Assumption 5 : } Q' = P \\
&\text{Assumption 6 : } \Delta I(P', P) + T_{B' \rightarrow P'} \geq 0.
\end{aligned} \tag{30}$$

These assumptions isolate a minimal class of interactions where merging behavior can be analytically characterized.

From the equation (10),

$$\begin{aligned}
T_{B' \rightarrow P'} = & \Delta H(P') + T_{Q' \rightarrow P'}^\dagger - T_{A' \rightarrow P'} \\
& + E_{\{A', B', Q'\} \rightarrow P'} - \alpha_1^{\{A', B', Q'\} \rightarrow P'}.
\end{aligned} \tag{31}$$

The term $E_{\{A', B', Q'\} \rightarrow P'}$ can be reduced to the single multiple transfer entropy:

Proposition 4 *From Assumption 2, 5,*

$$E_{\{A', B', Q'\} \rightarrow P'} = T_{\{A', B'\} \rightarrow P'}. \tag{32}$$

(proof) Appendix. C.

The Proposition 4 and Assumption 1 lead to:

$$T_{B' \rightarrow P'} = \Delta H(P') + T_{Q' \rightarrow P'}^\dagger - T_{A \rightarrow P} - \alpha_1^{\{A', B', Q'\} \rightarrow P'}. \tag{33}$$

The limit $H(P') \rightarrow \ln Q$ is encoded by introducing $\epsilon \geq 0$ such that,

$$\epsilon := \ln Q - \alpha_1^{\{A, Q\} \rightarrow P} - T_{A \rightarrow P}. \tag{34}$$

The limit $\epsilon \rightarrow 0$ encodes that the sum of the transferred information from A and internal noise, $\alpha_1^{\{A, Q\} \rightarrow P}$, reaches the maximum $\ln Q$. Plus, the additional small δ is considered:

$$\delta := \ln Q - \Delta H(P') - T_{Q' \rightarrow P'}^\dagger, \tag{35}$$

indicating,

$$T_{B' \rightarrow P'} = \epsilon - \delta, \tag{36}$$

from Eq. (33) and Assumption 3.

Before discussing the relations of ϵ and δ , one can deduce the lemma.

Lemma 1 *From Assumption 1, 2, 3, 5, and Proposition 4,*

$$H(P'' | P') = H(P' | P) + T_{B' \rightarrow P'}. \tag{37}$$

(proof) Appendix. D.

Then,

Proposition 5 *From Assumption 5, 6 and Lemma 1,*

$$\begin{aligned}
(a) \quad & H(P') \geq \ln Q - \epsilon \\
(b) \quad & 0 \leq \delta \leq \epsilon \\
(c) \quad & 0 \leq \Delta H(P') \leq \epsilon.
\end{aligned} \tag{38}$$

(proof) Appendix. E.

From Proposition 5, one obtains,

$$0 \leq T_{B' \rightarrow P'} \leq \epsilon. \quad (39)$$

Hence, in the limit $\epsilon \rightarrow 0$,

$$\begin{aligned} H(P') &\rightarrow \ln Q \\ \Delta H(P') &\rightarrow 0 \\ T_{Q' \rightarrow P'}^\dagger &\rightarrow \ln Q \\ T_{B' \rightarrow P'} &\rightarrow 0 \end{aligned} \quad (40)$$

From this limit, $T_{B' \rightarrow P'}$ can be expanded as:

$$T_{B' \rightarrow P'}(\epsilon) = \frac{\partial T_{B' \rightarrow P'}}{\partial \epsilon}(0)\epsilon + O(\epsilon^2), \quad (41)$$

assuming the differentiability of $T_{B' \rightarrow P'}$ near $\epsilon = 0$.

Given that Eq. (39),

$$\frac{\partial T_{B' \rightarrow P'}}{\partial \epsilon}(0) \geq 0, \quad (42)$$

indicating that $T_{B' \rightarrow P'}$ is an increasing function of ϵ near the point $\epsilon = 0$.

Since the Assumption 4 requires ϵ to decrease when $T_{A \rightarrow P}$ increases, $T_{B' \rightarrow P'}$ is regarded as a decreasing function of $T_{A \rightarrow P}$ near the point $\epsilon = 0$:

$$\frac{\partial T_{B' \rightarrow P'}}{\partial T_{A \rightarrow P}} = \frac{\frac{\partial T_{B' \rightarrow P'}}{\partial \epsilon}}{\frac{\partial T_{A \rightarrow P}}{\partial \epsilon}} \leq 0. \quad (43)$$

Equation (43) shows that, near the entropy-capacity limit of the merging node, an increase in the pre-existing circulating information flow suppresses the transfer entropy from the external source. Thus, in this limit, the internal circulation and the external input become effectively competitive. This provides an information-theoretic mechanism for robustness: a strong circulation protects the cycle by reducing the influence of perturbing inputs when the receiving node has limited entropy capacity. These arguments further indicate that the control variables α_3 and α_6 , introduced previously, must satisfy conditions leading to Eq. (43) near the entropy-capacity limit.

In communication theory [45, 46], this corresponds to the cross-talk effect in the two-senders-one-receiver multiple-access channels (MAC). The cross-talk [47, 48, 49] in MAC has been referred to as the difficulty of simultaneous communication with two senders, when the receiver's capacity for information processing is limited. Hence, the result of this section, Eq. (43) implies that the stronger the communication between components A and P is, the more difficult it becomes for component B to communicate with component P , if the individual entropy of component P is near the upper limit. This cross-talk effect can be resolved when considering a multiplex reaction mechanism as in Sec. 5.4.2.

Intuitively, a finite-capacity receiving component cannot accommodate arbitrarily many independent incoming influences. The present result renders this intuition quantitative at the level of directional information flow by deriving a specific bound, Eq. (39). This provides a minimal information-theoretic mechanism by which cyclic organization can contribute to robustness under finite entropy capacity.

5.4 Examples of merging information flows

The validity of Eq. (43) and Assumption 1 ~ 6 (Eq. (30)) are examined for specific dynamical systems.

5.4.1 Gaussian MAC Let the dynamics at time $t < t_1$ be,

$$\begin{aligned} A' &\sim N(0, \sigma_A) \\ B' &\sim N(0, \sigma_B) \\ P' &= A \\ Q' &= P, \end{aligned} \quad (44)$$

where $N(\mu, \sigma)$ is the Gaussian distribution with the mean μ and variance σ^2 , and σ_A and σ_B are assumed to be positive.

At time $t \geq t_1$,

$$\begin{aligned} A'' &\sim N(0, \sigma_A) \\ B'' &\sim N(0, \sigma_B) \\ P'' &= A' + B' \\ Q'' &= P', \end{aligned} \tag{45}$$

Since the distribution of P'' is the convolution of distributions of A' and B' , one obtains $P'' \sim N(0, \sqrt{\sigma_A^2 + \sigma_B^2})$. Therefore,

$$\begin{aligned} T_{A \rightarrow P} &= \ln(\sigma_A) + \frac{1}{2} \ln(2\pi e) \\ T_{B' \rightarrow P'} &= \frac{1}{2} \ln \frac{\sigma_A^2 + \sigma_B^2}{\sigma_A^2}, \end{aligned} \tag{46}$$

where e is the natural constant.

This yields,

$$\frac{dT_{B' \rightarrow P'}}{dT_{A \rightarrow P}} = \frac{\frac{dT_{B' \rightarrow P'}}{d\sigma_A}}{\frac{dT_{A \rightarrow P}}{d\sigma_A}} = -\frac{\sigma_B^2}{\sigma_A^2 + \sigma_B^2} < 0. \tag{47}$$

In this dynamics, the equation (43) is valid for the overall range. Also, it can be shown that the assumptions of Eq. (30) are valid.

5.4.2 A Multiplex Receiver On the other hand, most artificial networks, such as multi-source networks in communication theory, are designed to prevent the cross-talk effect for a credible communication. This is accomplished by modeling the multiplex receiver to process input signals independently. Therefore, the multiplex receiver is assigned two independent state-space components, $P = (P_1, P_2)$. Then, the dynamics is set as,

$$\begin{aligned} A' &\sim \text{Uniform}(N_{Z_A}) \\ B' &\sim \text{Uniform}(N_{Z_B}) \\ P'_1 &= A \\ P'_2 &= P_2 \\ Q'_1 &= P_1 \\ Q'_2 &= P_2, \end{aligned} \quad \text{for } t < t_1 \tag{48}$$

where $N_Z := \{1, 2, \dots, Z\}$ and $Z_A, Z_B \in \mathbb{N}$, and $\text{Uniform}(N_Z)$ is the uniform distribution for the set N_Z .

$$\begin{aligned} A'' &\sim \text{Uniform}(N_{Z_A}) \\ B'' &\sim \text{Uniform}(N_{Z_B}) \\ P''_1 &= A' \\ P''_2 &= B' \\ Q''_1 &= P'_1 \\ Q''_2 &= P'_2. \end{aligned} \quad \text{for } t \geq t_1 \tag{49}$$

As a result,

$$\begin{aligned} T_{A \rightarrow P} &= \ln(Z_A) \\ T_{B' \rightarrow P'} &= \ln(Z_B), \end{aligned} \tag{50}$$

and,

$$\frac{dT_{B' \rightarrow P'}}{dT_{A \rightarrow P}} = 0 \tag{51}$$

This represents a limiting case in which the cross-talk effect is eliminated, satisfying all assumptions of Eq. (30).

5.4.3 AND Gate Unlike the previous examples, the merge with AND gate violates some of the assumptions and does not satisfy Eq. (43) in general.

$$\begin{aligned} A' &\sim B(\frac{1}{2} + \rho_A) \\ B' &\sim B(\frac{1}{2} + \rho_B) \quad \text{for } t < t_1 \\ P' &= A \\ Q' &= P, \end{aligned} \tag{52}$$

where $B(\rho)$ is the binomial distribution with probability ρ and $-\frac{1}{2} \leq \rho_A, \rho_B \leq \frac{1}{2}$.

$$\begin{aligned} A'' &\sim B(\frac{1}{2} + \rho_A) \\ B'' &\sim B(\frac{1}{2} + \rho_B) \quad \text{for } t \geq t_1 \\ P'' &= \text{AND}(A', B') \\ Q'' &= P'. \end{aligned} \tag{53}$$

Here,

$$\begin{aligned} T_{A \rightarrow P} &= -(\frac{1}{2} + \rho_A) \ln(\frac{1}{2} + \rho_A) - (\frac{1}{2} - \rho_A) \ln(\frac{1}{2} - \rho_A) \\ T_{B' \rightarrow P'} &= -(\frac{1}{2} + \rho_A)(\frac{1}{2} + \rho_B) \ln(\frac{1}{2} + \rho_B) + (\frac{1}{2} - \rho_A)(\frac{1}{2} + \rho_B) \ln(\frac{1}{2} - \rho_A) \\ &\quad - (1 - (\frac{1}{2} + \rho_A)(\frac{1}{2} + \rho_B)) \ln(1 - (\frac{1}{2} + \rho_A)(\frac{1}{2} + \rho_B)). \end{aligned} \tag{54}$$

While,

$$\frac{\partial T_{B' \rightarrow P'}}{\partial \rho_A} = (\frac{1}{2} + \rho_B) \ln \frac{1 - (\frac{1}{2} + \rho_A)(\frac{1}{2} + \rho_B)}{(\frac{1}{2} - \rho_A)(\frac{1}{2} + \rho_B)} \geq 0, \tag{55}$$

$$\frac{\partial T_{A \rightarrow P}}{\partial \rho_A} = \ln \frac{\frac{1}{2} - \rho_A}{\frac{1}{2} + \rho_A} \begin{cases} < 0 & (\rho_A > 0) \\ > 0 & (\rho_A < 0) \\ = 0 & (\rho_A = 0) \end{cases} \tag{56}$$

Therefore,

$$\frac{dT_{B' \rightarrow P'}}{dT_{A \rightarrow P}} \begin{cases} \leq 0 & (\rho_A > 0) \\ \geq 0 & (\rho_A < 0) \end{cases} \tag{57}$$

The system satisfies the Assumption 2, 3, 5 clearly, however, Assumption 1 is invalid except the trivial case where $\rho_B = \frac{1}{2}$, and the validity of Assumption 4 depends on the value of ρ_A and ρ_B .

6 Example

To illustrate the mechanism derived in Sec. 5, we consider a stochastic network with discrete state space: a cycle $P \rightarrow A1 \rightarrow \dots \rightarrow A4 \rightarrow P$ and an external node (Ext). The merging node P represents a regulatory unit (e.g. a gene) that integrates inputs from $A4$ and (Ext). Each node has a binary state, $\Gamma_{A1}, \dots, \Gamma_{A4}, \Gamma_P, \Gamma_{Ext} \in \{0, 1\}$, and the update rule is given by a sigmoid response [50] to additive inputs,

$$\mathbb{P}(P' = 1) = \sigma(\beta_A \Gamma_{A4} + \beta_{Ext} \Gamma_{Ext} + \kappa), \tag{58}$$

where $\sigma(x) = 1/(1 + e^{-x})$ and $\kappa > 0$.

The update rule follows a standard logistic activation form, widely used in stochastic neural dynamics [51, 52] and thermodynamic models of gene regulation [53, 54], where the probability of activation depends on the combined influence of multiple inputs. In regimes where regulatory interactions are non-cooperative, thermodynamic models yield additive contributions from individual inputs, leading to sigmoidal activation functions of Hill type [53, 54]. In such models, the additive structure arises from independent contributions of inputs, which combine linearly in the effective activation argument.

The remaining nodes in the cycle are modeled as delay-copy units, such as $\Gamma'_{A2} = \Gamma_{A1}$ or $\Gamma'_{A1} = \Gamma_P$, ensuring consistency with Assumption 5 in Sec. 5. Within this construction, the cycle

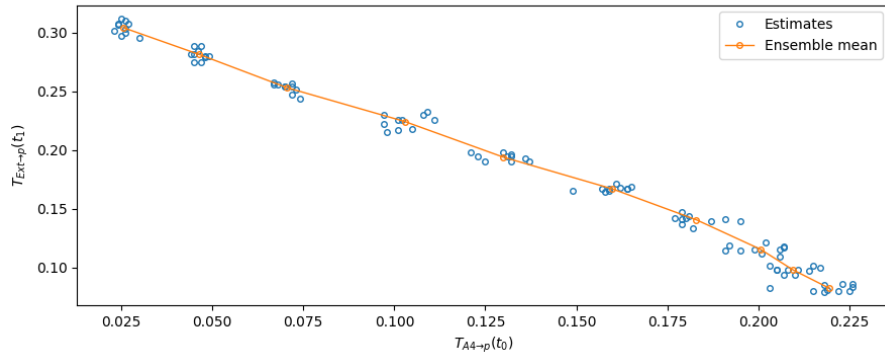


Figure 3: Simulation results for competing information flows: $T_{A4 \rightarrow P}(t_0)$ versus $T_{Ext \rightarrow P}(t_1)$. The cycle $P \rightarrow A1 \rightarrow \dots \rightarrow A4 \rightarrow P$ is isolated at time $t = t_0$, and node P begins to interact with the external source (Ext) at $t = t_1$.

represents a persistent internal information circulation, while the node P acts as a finite-capacity processor where internal and external information flows compete.

To evaluate the resulting dynamics, we simulate an ensemble of independent realizations and estimate the relevant information-theoretic quantities using a simple binning method. The binary network is constructed with the cycle $P \rightarrow A1 \rightarrow \dots \rightarrow A4 \rightarrow P$ and the external source (Ext), and the component P and (Ext) begin to interact after time $t = t_1$.

Simulation results (Fig. 3) [44] show that increasing the internal circulating flow $T_{A4 \rightarrow P}(t_0)$ suppresses the external input $T_{Ext \rightarrow P}(t_1)$, quantitatively confirming the predicted relation (Eq. (43)).

7 Conclusion

This work establishes information-dynamical descriptions that demonstrate: the circulation of transfer entropy and reversed transfer entropy constitutes a necessary structural condition for nontrivial dynamics in finite systems; and pre-existing information circulation can restrict subsequent incoming transfer entropy when a shared receiving node is close to its local entropy capacity. In this sense, robustness is not imposed externally but emerges from the compatibility between cyclic information flow and finite local entropy capacity. This suppression mechanism corresponds to a cross-talk effect, in which limited entropy capacity constrains simultaneous information transfer.

This perspective on robustness aligns with observations in gene regulatory networks, where cyclic structures are associated with robustness against external noise [6, 7]. Consequently, the quantitative framework developed in this work enables detailed analysis of biological networks to disentangle distinct origins of robustness.

The current analysis relies on several assumptions introduced in Sec. 2 and Eq. (30). While these assumptions facilitate tractable analysis of interacting information flows and yield baseline cases, their necessity and generality remain to be examined. Further work is required to relax these assumptions, explore a wider class of merging mechanisms, and investigate additional properties of information circulation, including its temporal structure and interactions between multiple cycles.

Furthermore, the present analysis assumes that the directed structure of transfer entropy interactions remains sufficiently stable over the analyzed time interval. Extending the framework to adaptive networks, in which the interaction topology and effective information flow structure evolve dynamically, remains an important direction for future research.

Data Availability Statement

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.20024405> (Version v0.1.3).

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A A Counterexample to the Assumption of Section 2.4

Let the random variables u_1 , u_2 , v represent the state of three components in discrete time having the dynamics:

$$u'_1, u'_2 \sim B\left(\frac{1}{2}\right) \quad v' = u_1 \oplus u_2, \quad (59)$$

where $B(p)$ is the binomial distribution with probability p , and the operation $\oplus$ represents the binary operation XOR.

Then, $H(v'|v) = H(v'|v, u_1) = H(v'|v, u_2) = \ln 2$, resulting in $T_{u_1 \rightarrow v} = T_{u_2 \rightarrow v} = 0$. Nonetheless, for $\mathbf{u} = (u_1, u_2)$, $T_{\mathbf{u} \rightarrow v} = H(v'|v) - H(v'|v, u_1, u_2) = \ln 2$.

B The Proof of Proposition 3

From the intrinsic method of Ref. [26], $\alpha_{3456}^{\Gamma_j \rightarrow \Gamma_i} := -\alpha_3^{\Gamma_j \rightarrow \Gamma_i} - \alpha_4^{\Gamma_j} + \alpha_5^{\Gamma_i, \Gamma_j} + \alpha_6^{\Gamma_j \rightarrow \Gamma_i}$ can be expressed as:

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{y} - \mathbf{A}^{-1}\mathbf{b}, \quad (60)$$

where $\mathbf{y}, \mathbf{x}, \mathbf{b} \in \mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$ such that,

$$\begin{aligned} (\mathbf{y})_l &= \alpha_3^{\Gamma_i \rightarrow \Gamma_{j_l}} - \alpha_6^{\Gamma_i \rightarrow \Gamma_{j_l}} - \Delta\alpha_2^{\Gamma_i, \Gamma_{j_l}} \\ (\mathbf{x})_l &= \alpha_{3456}^{\Gamma_{j_l} \rightarrow \Gamma_i} \\ (\mathbf{b})_l &= \Delta E_{K_i \rightarrow \Gamma_i} - \Delta\alpha_1^{K_i \rightarrow \Gamma_i} \\ (\mathbf{A})_{lm} &= -1 + \delta_{lm}, \end{aligned} \quad (61)$$

Here, the component i 's neighbor set is denoted as $K_i = \{\Gamma_{j_1}, \dots, \Gamma_{j_n}\}$ with $n > 1$.

Therefore,

($i \neq p$ case)

$$\begin{aligned} \alpha_{3456}^{\Gamma_{i-1} \rightarrow \Gamma_i} &= -\alpha_3^{\Gamma_i \rightarrow \Gamma_{i+1}} + \alpha_6^{\Gamma_i \rightarrow \Gamma_{i+1}} + \Delta E_{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} \\ &\quad - \Delta\alpha_1^{\Gamma_i} + \Delta\alpha_2^{\Gamma_i, \Gamma_{i+1}} \\ \alpha_{3456}^{\Gamma_{i+1} \rightarrow \Gamma_i} &= -\alpha_3^{\Gamma_i \rightarrow \Gamma_{i-1}} + \alpha_6^{\Gamma_i \rightarrow \Gamma_{i-1}} + \Delta E_{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} \\ &\quad - \Delta\alpha_1^{\Gamma_i} + \Delta\alpha_2^{\Gamma_i, \Gamma_{i-1}}. \end{aligned} \quad (62)$$

($i = p$ case)

$$\begin{pmatrix} \alpha_{3456}^{\Gamma_{p-1} \rightarrow \Gamma_p} \\ \alpha_{3456}^{\Gamma_{p+1} \rightarrow \Gamma_p} \\ \alpha_{3456}^{Ext \rightarrow \Gamma_p} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} +1 & -1 & -1 \\ -1 & +1 & -1 \\ -1 & -1 & +1 \end{pmatrix} \mathbf{y} + \frac{1}{2} \mathbf{b}, \quad (63)$$

where,

$$\begin{aligned} \mathbf{y} &= \begin{pmatrix} \alpha_3^{\Gamma_p \rightarrow \Gamma_{p-1}} - \alpha_6^{\Gamma_p \rightarrow \Gamma_{p-1}} - \Delta\alpha_2^{\Gamma_p, \Gamma_{p-1}} \\ \alpha_3^{\Gamma_p \rightarrow \Gamma_{p+1}} - \alpha_6^{\Gamma_p \rightarrow \Gamma_{p+1}} - \Delta\alpha_2^{\Gamma_p, \Gamma_{p+1}} \\ \alpha_3^{\Gamma_p \rightarrow Ext} - \alpha_6^{\Gamma_p \rightarrow Ext} - \Delta\alpha_2^{\Gamma_p, Ext} \end{pmatrix}, \\ \mathbf{b} &= [\Delta E_{\{\Gamma_{p-1}, \Gamma_{p+1}, Ext\} \rightarrow \Gamma_p} - \Delta\alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, Ext\} \rightarrow \Gamma_p}] \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \end{aligned} \quad (64)$$

Substituting these relations into the dynamical equations (Eq. (10)) and imposing the blocking flows condition, $T_{\Gamma_i \rightarrow \Gamma_{i-1}}(t) = 0$, $T_{\Gamma_i \rightarrow \Gamma_{i+1}}^\dagger(t) = 0$, $T_{\Gamma_p \rightarrow Ext}(t) = 0$, $T_{Ext \rightarrow \Gamma_p}^\dagger(t) = 0$ for $\forall t, i$, yields,

$$\begin{aligned} \Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}} &= T_{\Gamma_{i-1} \rightarrow \Gamma_i} - T_{\Gamma_i \rightarrow \Gamma_{i+1}} - E_{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} + \alpha_1^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} \\ &\quad - \alpha_2^{\Gamma_i, \Gamma_{i+1}} - \alpha_3^{\Gamma_i \rightarrow \Gamma_{i+1}} \quad (i \neq p) \\ \Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}}^\dagger &= 0 \\ \Delta T_{\Gamma_i \rightarrow \Gamma_{i-1}} &= 0 \\ \Delta T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger &= T_{\Gamma_{i-1} \rightarrow \Gamma_{i-2}}^\dagger - T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger - E'_{\{\Gamma_{i-2}, \Gamma_i\} \rightarrow \Gamma_{i-1}} + \alpha_1^{\{\Gamma_{i-2}, \Gamma_i\} \rightarrow \Gamma_{i-1}} \\ &\quad - \alpha_2^{\Gamma_{i-1}, \Gamma_i} - \alpha_6^{\Gamma_{i-1} \rightarrow \Gamma_{i-2}} \quad (i \neq p+1) \\ \Delta T_{\Gamma_p \rightarrow \Gamma_{p+1}} &= T_{\Gamma_{p-1} \rightarrow \Gamma_p} + T_{Ext \rightarrow \Gamma_p} - T_{\Gamma_p \rightarrow \Gamma_{p+1}} - E_{\{\Gamma_{p-1}, \Gamma_{p+1}, Ext\} \rightarrow \Gamma_p} \\ &\quad + \alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, Ext\} \rightarrow \Gamma_p} - \alpha_2^{\Gamma_p, \Gamma_{p+1}} - \alpha_3^{\Gamma_p \rightarrow \Gamma_{p+1}} \\ \Delta T_{\Gamma_{p+1} \rightarrow \Gamma_p}^\dagger &= \alpha_1^{\{\Gamma_p\} \rightarrow Ext} - \alpha_3^{Ext \rightarrow \Gamma_p} + T_{\Gamma_p \rightarrow \Gamma_{p-1}}^\dagger - T_{\Gamma_{p+1} \rightarrow \Gamma_p}^\dagger - \alpha_6^{\Gamma_p \rightarrow \Gamma_{p-1}} \\ &\quad - E'_{\{\Gamma_{p-1}, \Gamma_{p+1}, Ext\} \rightarrow \Gamma_p} + \alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, Ext\} \rightarrow \Gamma_p} - \alpha_2^{\Gamma_p, \Gamma_{p-1}} \\ \Delta T_{Ext \rightarrow \Gamma_p} &= \alpha_1^{\{\Gamma_p\} \rightarrow Ext} - T_{Ext \rightarrow \Gamma_p} - \alpha_3^{Ext \rightarrow \Gamma_p} \\ \Delta T_{\Gamma_p \rightarrow Ext}^\dagger &= -\alpha_{3456}^{\Gamma_p \rightarrow Ext}, \end{aligned} \quad (65)$$

where E' denotes for E evaluated at the following time step.

As Ref. [26], it is required to analyze near the zero-control limit for understanding the impact of merge. Therefore,

$$\begin{aligned}
\alpha_1^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} &= \alpha_2^{\Gamma_i, \Gamma_{i+1}} = 0 \quad (i \neq p) \\
\alpha_1'^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} &= \alpha_2'^{\Gamma_i, \Gamma_{i+1}} = 0 \quad (i \neq p) \\
\alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} &= \alpha_1'^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} = \alpha_2^{\Gamma_i, \Gamma_{i+1}} = \alpha_2'^{\Gamma_i, \Gamma_{i+1}} = 0 \\
\alpha_3^{\Gamma_i \rightarrow \Gamma_{i+1}} &= 0 \quad (i \neq p-1) \\
\alpha_6^{\Gamma_{i-1} \rightarrow \Gamma_{i-2}} &= 0 \quad (i \neq p+1),
\end{aligned} \tag{66}$$

retaining only the control variables associated with the newly opened channel between Ext and p . In this limit, $\Delta E = 0$, as the high-order control variables also vanish. Therefore, the values of $E_{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i}$ and $E'_{\{\Gamma_{i-2}, \Gamma_i\} \rightarrow \Gamma_{i-1}}$ are assumed to be zero from sufficient initial conditions.

This results in the Eq. (25).

■

The equation (25) requires further specification. Firstly, $\alpha_{3456}^{\Gamma_p \rightarrow \Gamma_{Ext}}$ cannot be replaced by α_3, α_6 by the intrinsic method introduced in Ref. [26], since the component Ext has only one neighbor.

Therefore, $\alpha_{3456}^{\Gamma_p \rightarrow \Gamma_{Ext}}$ has to be determined from its definition,

$\alpha_{3456}^{\Gamma_p \rightarrow \Gamma_{Ext}} = -\alpha_3^{\Gamma_p \rightarrow \Gamma_{Ext}} - \alpha_4^{\Gamma_p} + \alpha_5^{\Gamma_p, \Gamma_{Ext}} + \alpha_6^{\Gamma_p \rightarrow \Gamma_{Ext}}$. Secondly, defining the dynamic characteristics of the component Ext can specify the terms, $\alpha_5^{\Gamma_p, \Gamma_{Ext}}$ and $\alpha_1^{\{\Gamma_p\} \rightarrow \Gamma_{Ext}}$. For instance, $\alpha_5^{\Gamma_p, \Gamma_{Ext}} = 0$ at all times and $\alpha_1^{\{\Gamma_p\} \rightarrow \Gamma_{Ext}} = 0$ at $t < t_1$ if the component Ext is regarded as a source of white noise.

On the other hand, if $E_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} = 0$ permanently, there is no stationary state with non-zero $T_{\Gamma_{Ext} \rightarrow \Gamma_p}, T_{\Gamma_p \rightarrow \Gamma_{Ext}}^\dagger$. This shows that multivariate effects are essential for the existence of a stationary state with nonzero interaction with the external source.

Such transient effects are expected to appear in explicit dynamical realizations of the merging process. The transient behaviors after time $t = t_1$ and the way to approach another stationary state were demonstrated numerically in Ref. [44].

C The Proof of Proposition 4

By the definition of E ,

$$\begin{aligned}
E_{\{A, B, Q\} \rightarrow P} &= [T_{\{A, B\} \rightarrow \{P\}} - T_{\{A, B\} \rightarrow \{P\}}^\dagger] + [T_{\{A, Q\} \rightarrow \{P\}} - T_{\{A, Q\} \rightarrow \{P\}}^\dagger] \\
&\quad + [T_{\{B, Q\} \rightarrow \{P\}} - T_{\{B, Q\} \rightarrow \{P\}}^\dagger] - [T_{\{A, B, Q\} \rightarrow \{P\}} - T_{\{A, B, Q\} \rightarrow \{P\}}^\dagger].
\end{aligned} \tag{67}$$

From Assumption 5, one obtains,

$$\begin{aligned}
T_{\{A, Q\} \rightarrow \{P\}}^\dagger &= I(\{A', Q', P\} | P') = I(A', Q' | P') - I(A', Q' | PP') \\
&= I(A', P | P') \quad (\because \text{Assumption 5: } Q' = P) \\
&= T_{A \rightarrow p}^\dagger = 0 \quad (\because \text{blocking flows condition}),
\end{aligned} \tag{68}$$

and $T_{\{B, Q\} \rightarrow \{P\}}^\dagger = 0$ in the similar manner. In addition,

$$\begin{aligned}
T_{\{A, B, Q\} \rightarrow \{P\}}^\dagger &= I(\{A', B', Q', P\} | P') = I(\{A', B', P\} | P') - I(\{A', B', P\} | P'Q') \\
&= I(\{A', B', P\} | P') \quad (\because \text{Assumption 5: } Q' = P) \\
&= T_{\{A, B\} \rightarrow \{P\}}^\dagger.
\end{aligned} \tag{69}$$

The $T_{\{A, B, Q\} \rightarrow \{P\}}$ is estimated with the Assumption 2:

$$\begin{aligned}
T_{\{A, B, Q\} \rightarrow \{P\}} &= I(\{A, B, Q, P'\} | P) = I(\{A, Q, P'\} | P) - I(\{A, Q, P'\} | PB) \\
&= T_{\{A, Q\} \rightarrow \{P\}} - I(Q, P' | PB) + I(Q, P' | PAB) \\
&= T_{\{A, Q\} \rightarrow \{P\}} + [I(Q, P' | P) - I(Q, P' | PB)] + I(Q, P' | PAB) \\
&\quad (\because I(Q, P' | P) = T_{Q \rightarrow P} = 0) \\
&= T_{\{A, Q\} \rightarrow \{P\}} + T_{\{B, Q\} \rightarrow \{P\}} \\
&\quad (\because I(Q, P' | PAB) = H(P' | PAB) - H(P' | PABQ) = 0).
\end{aligned} \tag{70}$$

By summing up above results, the Eq. (67) becomes,

$$E_{\{A,B,Q\} \rightarrow P} = T_{\{A,B\} \rightarrow \{P\}}. \quad (71)$$

■

D The Proof of Lemma 1

By the Assumption 2 and 5,

$$\begin{aligned} \alpha_1^{\{A,Q\} \rightarrow P} &= H(P'|PAQ) - H(P|P'A'Q') = H(P'|PA) \\ \alpha_1^{\{A',B',Q'\} \rightarrow P'} &= H(P''|P'A'B'Q') - H(P'|P''A''B''Q'') = H(P''|P'A'B'). \end{aligned} \quad (72)$$

Since,

$$\begin{aligned} H(P'|P) &= T_{A \rightarrow P} + H(P'|PA) \\ &= T_{A \rightarrow P} + \alpha_1^{\{A,Q\} \rightarrow P}, \end{aligned} \quad (73)$$

$$\begin{aligned} H(P''|P') &= T_{B' \rightarrow P'} + T_{A' \rightarrow P'} - T_{\{A',B'\} \rightarrow \{P'\}} + H(P''|P'A'B') \\ &= T_{B' \rightarrow P'} + T_{A \rightarrow P} + H(P''|P'A'B') (\because \text{Assumption1}) \\ &= T_{B' \rightarrow P'} + T_{A \rightarrow P} + H(P'|PA) (\because \text{Assumption3}) \\ &= T_{B' \rightarrow P'} + H(P'|P). \end{aligned} \quad (74)$$

■

E The Proof of Proposition 5

(a)

From Eq. (73),

$$\begin{aligned} H(P'|P) &= T_{A \rightarrow P} + \alpha_1^{\{A,Q\} \rightarrow P} \\ &= \ln Q - \epsilon, \end{aligned} \quad (75)$$

one obtains,

$$H(P') \geq H(P'|P) = \ln Q - \epsilon. \quad (76)$$

(b)

From Assumption 5,

$$T_{Q' \rightarrow P'}^\dagger = H(Q'|P') - H(Q'|PP') = H(P|P') = -\Delta H(P) + H(P'|P). \quad (77)$$

As a result,

$$H(P''|P') = \ln Q - \delta. \quad (78)$$

Since $H(P''|P') \leq H(P'') \leq \ln Q$, one obtains $0 \leq \delta$.

Here, $\delta \leq \epsilon$ is derived from the non-negativity of $T_{B' \rightarrow P'}$.

(c)

From Assumption 6 and Lemma 1,

$$\Delta I(P', P) = \Delta H(P') - \Delta H(P'|P) = \Delta H(P') - T_{B' \rightarrow P'} \geq -T_{B' \rightarrow P'}. \quad (79)$$

Therefore,

$$\Delta H(P') \geq 0. \quad (80)$$

Then,

$$\ln Q \geq H(P'') \geq H(P') \geq \ln Q - \epsilon. \quad (81)$$

Hence,

$$\Delta H(P') \leq \ln Q - (\ln Q - \epsilon) = \epsilon. \quad (82)$$

■